Assumptions

- A local authority may be recorded before signing a contract with Clear Away, because it may still be negotiating the contract.
- A waste type may be recorded before it is included in any contract.
- A bin type may be recorded before it is made available under any contract.
- An owner is recorded only if they currently own at least one property in the system.
- A truck may exist in the system before being assigned to any collection run.
- A contract waste type coverage may exist before any collection runs are made under it.
- A bin may exist in the system before being collected.

Logical Model

WASTE_TYPE is implemented as a lookup table itself rather than a check constraint because the case study states the company would like to easily add new types of waste. A lookup table allows new waste types to be added by inserting new rows.

bin_replacement_reason is nullable because not all bins are replacements. A new bin supplied to a property for the first time has no replacement reason.

Surrogate Choice:

- BIN_TYPE: because we want to get rid of the waste_type_id appear twice in CONTRACT_BIN, where it has 2 waste_type_id, making bin_type_id (waste_type_id, bin_size), so CONTRACT_BIN will use this bin_type_id but still have enough Primary keys.
  waste_type_id appears in CONTRACT_BIN as part of the composite FK to CONTRACT_WASTE. Although waste_type_id is already captured through bin_type_id, I think its better to maintain referential integrity with CONTRACT_WASTE's composite primary key.
- CONTRACT_BIN: it originally holds 5 attributes but through bin_type_id, it still holds 3, make it minimal using cbt_id, and this will also avoid the update anomaly risk, if any natural key changes, its child change too.
- COLLECTION: because, COLLECTION_BIN references it and COLLECTION_BIN also identified by BIN which alseo have contract_no and waste_type_id, this will avoid duplicate in COLLECTION_BIN and make it less complex.
- PROPERTY: its because authority_name is text-based natural key, so any update might required across STREET, PROPERTY, and PROPERTY_OWNER, making a update anomaly risk. A prop_id will void this. And make BIN look cleaner.